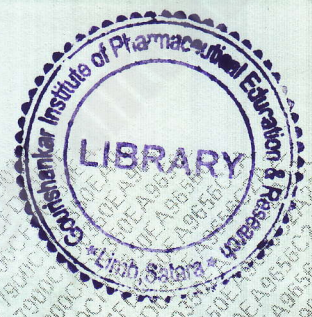




DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Summer 2022

Course: B. Pharmacy

Sem: VI

Subject Name: Biopharmaceutics & Pharmacokinetics

Subject Code: BP604T

Max. Marks: 75

Date: 22/07/2022

Duration: 3.45 Hrs.

Instructions –

1. All questions are compulsory
2. Answers to MCQs should be written in full sentences
3. Draw diagrams / figures wherever necessary
4. Figures to right indicate full marks

Q. 1. Multiple Choice Questions (MCQs) = 20 x 1 = 20 (All the questions are compulsory)

i) Drug absorption by passive diffusion is

- | | |
|------------------------|---------------------|
| a) Non ionic diffusion | b) Energy dependent |
| c) Uphill transport | d) Saturable |

ii) Acidic drugs mainly bind to

- | | |
|---------------------------------|--------------|
| a) α - Acid glycoprotein | b) Antigens |
| c) human serum albumin | d) Vitamin A |

iii) In general, the presence of food in the gastrointestinal tract (GIT) reduces the rate and extent of drug absorption due to

- a) Enhanced presystemic drug elimination
- b) High viscosity of the GIT contents
- c) Increased adsorption of the drug on the GI contents
- d) Increased dissolution of the drug in the GI contents

iv) Calculate the volume of distribution of drug, when 300 mg dose is administered showed the initial blood drug concentration of 30 microgram/mL?

- | | | | |
|--------|---------|----------|---------|
| a) 9 L | b) 10 L | c) 100 L | d) 90 L |
|--------|---------|----------|---------|

v) Gaseous and volatile substances are excreted through one of the following routes.

- | | |
|-----------|--------------|
| a) Dermal | b) Pulmonary |
| c) Renal | d) Salivary |

vi) Which is NOT a marker for renal drug excretion?

- a) Creatinine b) Glucose
c) Inulin d) Probencid

vii) One of the following is the phase II drug biotransformation reaction?

- a) Acetylation b) Deamination
c) Hydrolysis d) Reduction

viii) Conjugation with Glutathione useful for

- a) water soluble metabolite formation b) detoxification
c) enhancing biological half life d) Reduction

ix) What does the word "open" mean in the one compartment open model?

- a) Unidirectional input and output b) The drug readily mixes with the blood
c) The drug easily enters d) Easy absorption

x) In one compartment open model drug disposition, the drug is eliminated by one of the following patterns:

- a) Biexponentially b) Nonexponentially
c) Triexponentially d) Monoexponentially

xi) In one compartment open model drug disposition, the assigned one compartment is:

- a) Body b) Blood
c) Gastrointestinal tract d) Liver

xii) In which of the model peripheral compartments are connected to a central compartment in series?

- a) Catenary model b) Physiologic model
c) Compartment model d) Mammillary model

xiii) Steady state plasma concentration (C_{ss}) depends on following factors except

- a) Clearance b) Dosing interval c) Dose d) Elimination half life

xiv) Total systemic clearance in two compartment open model is calculated by

- a) hybrid first order constants for slow elimination phase and apparent volume of distribution
b) hybrid first order constants for slow elimination phase and Clearance
c) hybrid first order constants for slow elimination phase and AUC
d) hybrid first order constants for slow elimination phase and rapid distribution phase

xv) The dosing interval depends on one of the following

- a) Apparent volume of distribution b) AUC
- c) Clearance d) Plasma Elimination half life

xvi) The loading dose of a drug is based upon the desired plasma drug concentration and

- a) Time taken for complete elimination
- b) Percentage of drug excreted unchanged in urine
- c) Percentage of drug bound to plasma protein
- d) Apparent volume of distribution

xvii) One of the following statements is correct with respect to non-linear pharmacokinetics.

- a) First order b) First order followed by zero order
- c) Pseudo first order d) Zero order

xviii) Xenobiotic means

- a) foreign to body b) Produced by bacteria
- c) produced by xerophytes d) synthesized in the body

xix) The volume of distribution of a drug is

- a) an expression of total body volume
- b) a measure of total fluid volume
- c) a relationship between the total amount of drug in body and plasma concentration of drug
- d) proportional to bioavailability of the drug

xx) What is the reason of complicated penetration of some drugs through Blood brain barrier?

- a) High lipid solubility of drug
- b) High endocytosis degree in a brain capillary
- c) Absence of pores in the brain capillary endothelium
- d) Meningitis

Q. 2. Long Answers) = 2 x 10 = 20 (Answer 2 out of 3)

- i) Describe One compartment open model. Deduce the monoexponential equation of disposition of drugs. Illustrate assessment of Pharmacokinetic parameters after IV bolus administration of drug for One compartment open model.
- ii) Define absorption. List factors influencing absorption of drugs. Discuss physicochemical factors in detail.

iii) Define Bioavailability and bioequivalence. Explain Pharmacokinetic & Pharmacodynamic Methods of assessing bioavailability.

Q. 3. Short Answers = 7 x 5 = 35 (Answer 7 out of 9)

- i) List USP *In vitro* dissolution test apparatus and illustrate any 4 apparatus.
- ii) Compare active and passive transport of drug absorption mechanism.
- iii) Differentiate between loading dose and maintenance dose.
- iv) Describe physiological models.
- v) Illustrate Phase I metabolism reactions with example.
- vi) Explain Wagner Nelson method for estimation of K_a .
- vii) Illustrate factors causing Non-linearity in pharmacokinetics with example of drugs.
- viii) Explain protein binding of drugs.
- ix) Explain mechanism of renal excretion. State the equation of renal clearance.
List factors affecting renal excretion of drugs.

-----END OF THE PAPER-----

Course- B. Pharmacy

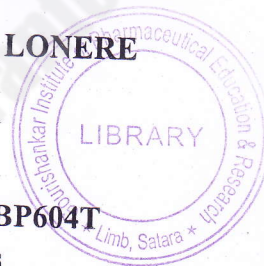
Subject Name- Biopharmaceutics & Pharmacokinetics

Max. Mark- 75

Sem-VI

Subject code- BP604T

Duration-3 hrs



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) Give characteristics of an active transport mechanism.
- ii) What is enterohepatic cycling of a drug?
- iii) List factors affecting protein binding of drugs.
- iv) Give benefits of drug clearance.
- v) Comment on particle size as a tool to enhance bioavailability of drugs.
- vi) What is apparent volume of distribution.
- vii) Differentiate between transcellular & paracellular transport.
- viii) Write importance of urinary excretion data.
- ix) Enlist patient related factors of GI absorption of drugs.
- x) Explain concept of clearance.

Q. 2. Long Answers (Answer 2 out of 3) (5x10) = 20

- i) Define drug absorption. Discuss physiochemical factors influencing GI absorption of drugs.
- ii) Explain in detail one compartment open model- intravenous bolus administration.
- iii) Discuss the methods used for enhancing bioavailability through enhancement of drug solubility or dissolution rate.

Q. 3. Short Answers (Answer 7 out of 9) (7x5) = 35

- i) Explain various theories of drug dissolution.
- ii) Write a note on non-renal routes of drug excretion.
- iii) Explain passive diffusion & facilitated diffusion mechanism of drug transport.
- iv) What is IVIVC? Give its significance.
- v) Explain sigma minus method for determination of pharmacokinetic parameters.
- vi) Write a note on protein binding of drugs.
- vii) Discuss about pH partition hypothesis.
- viii) Explain method of residuals.
- ix) Explain two compartment open model for IV bolus administration.

Course- B. Pharmacy

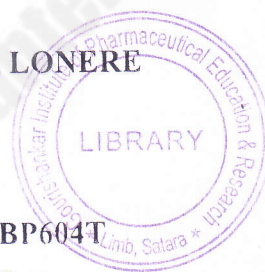
Subject Name- Biopharmaceutics & Pharmacokinetics

Max. Mark- 75

Sem-VI

Subject code- BP604T

Duration-3 hrs



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

(10 x 2) = 20

Q. 1. Objective Type Questions (Answer all the questions)

- i) Define i) Absorption ii) Distribution.
- ii) Define Volume of Distribution
- iii) Why there is need of Biotransformation.
- iv) Enlist Dissolution test apparatus according to USP.
- v) Define i) C_{max} ii) T_{max}.
- vi) Compare between Plasma protein binding & Tissue binding.
- vii) Draw a presentation for One compartment first order absorption model for extravascular route of administration.
- viii) Explain mixed order kinetics.
- ix) Draw a presentation for Two compartment open model for intravenous Infusion.
- x) Explain Zero order Kinetics.

Q. 2. Long Answers (Answer 2 out of 3)

(5x10) = 20

- i) Explain nonlinearity, methods to determine nonlinearity & causes of nonlinearity.
- ii) Explain one compartment open model for intravenous infusion.
- iii) Explain factor influencing drug absorption of drug.

Q. 3. Short Answers (Answer 7 out of 9)

(7x5) = 35

- i) Explain methods of measurement of Bioavailability.
- ii) Explain mechanism of drug absorption.
- iii) Explain chemical pathways of drug biotransformation.
- iv) Explain one compartment open model for intravenous injection.

- v) Explain Mammillary model
- vi) Write a note on protein binding of drugs.
- vii) Explain kinetics of multiple dosing.
- viii) Explain renal clearance with factors affecting renal clearance.
- ix) Explain factor affecting distribution of drug.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary End Semester Examination – Winter 2023

Course : B. Pharmacy

Date:- 05/01/24

Sem : VI

Subject Name –Biopharmaceutics and Pharmacokinetics

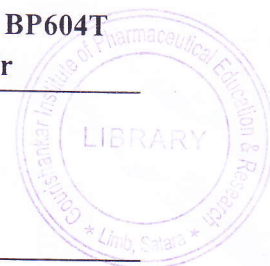
Subject Code : BP604T

Max Marks: 75

Duration : 3 Hr

Instructions:

1. All Questions Are Compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks



Q.1 Objective Type Questions (All Questions are compulsory)

(10 x 2) = 20

- i. Enlist different mechanisms of drug absorption.
- ii. Define volume of distribution with example.
- iii. Enlist non oral extra vascular route of drug administration.
- iv. Enlist different routes of drug elimination.
- v. Enlist factors affecting renal excretion.
- vi. Discuss in brief about Phase I biotransformation reactions of metabolism
- vii. Write the formula for Renal clearance (Cl_R) & Total Clearance (Cl_T).
- viii. Explain steady state drug level.
- ix. Define dosage regimen.
- x. Write Michaelis-menton equation.

Q.2 Long Answer Question (Answer any 2 out of 3)

(2 x 10) = 20

- i. Define Bioavailability and Bioequivalence. Explain the objectives of bioavailability studies. Describe in detail methods used for determination of bioavailability.
- ii. Discuss the concept of drug absorption. Enlist factors influencing absorption of drugs. Discuss pharmaceutical factors in detail.
- iii. Explain in detail two compartment open model. Illustrate assessment of Pharmacokinetic parameters after IV bolus administration of drug for two compartment open model.

Q.3 Short Answer Questions (Solve any 7 out of 9)

(7 x 5) = 35

- i. Explain any 2 Mechanisms of drug absorption through GIT.
- ii. Explain the concept of protein binding and illustrate any 4 factors affecting drug distribution.
- iii. Illustrate Non renal routes of drug excretion of drugs.
- iv. Explain Wagner Nelson method for estimation of K_a .
- v. Describe compartment models.
- vi. Explain the concept of loading and maintenance dose.
- vii. Explain the factors causing Non-linearity in pharmacokinetics with suitable examples.
- viii. Enlist USP *In vitro* dissolution test apparatus and illustrate any 4 apparatus.
- ix. Describe physiological models.

===== End Of Paper =====

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Summer Examination – 2024

Course: B. Pharmacy

Subject Name: BIOPHARMACEUTICS AND PHARMACOKINETICS

Semester: VI

Subject Code: BP604T

Max Marks: 75

Date: 20-06-2024

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1. Objective Type Questions (Answer All the Questions) (10 X 2) = 20

- i) Explain in short about an active transport of drug.
- ii) Write in short about non renal route of Excretion.
- iii) What is loading & maintenance dose?
- iv) Define bioavailability. What are the objectives of bioavailability studies?
- v) Define and explain renal clearance.
- vi) Explain in short about a passive transport of drug.
- vii) What are the advantages of administering a drug by constant rate i.v. infusion over oral administration?
- viii) Enlist the components of blood to which drug binds.
- ix) What is AUC & $t_{1/2}$?
- x) Give clinical significance of protein binding of drugs.

Q.2. Long Answers (Answer 2 out of 3) (10 X 2) = 20

- i) Enlist various factors affecting absorption and explain in detail pharmaceutical factors affecting absorption of drug from GIT
- ii) Write a detail note on kinetics of protein binding.
- iii) Define distribution of drug. Discuss various factors affecting distribution of drug.

Q.3. Short Answers (Answer 7 out of 9) (5 X 7) = 35

- i) Volume of distribution and its importance.
- ii) Write a note on causes of Non-linear pharmacokinetics.
- iii) What are various approaches used to enhance bioavailability of drug from its dosage form.
- iv) Write a note on In-Vitro and In-Vivo co-relation.
- v) What are various sites of drug metabolism in the body?
- vi) Write detailed note on Blood Brain Barrier.
- vii) Explain the design for one compartmental open model for the zero-order i.v. infusion.
- viii) Explain Michaelis Menten equation and determination of V_{max} and K_m .
- ix) Why drugs are better absorbed from small intestine? Explain.

*** END OF THE PAPER ***